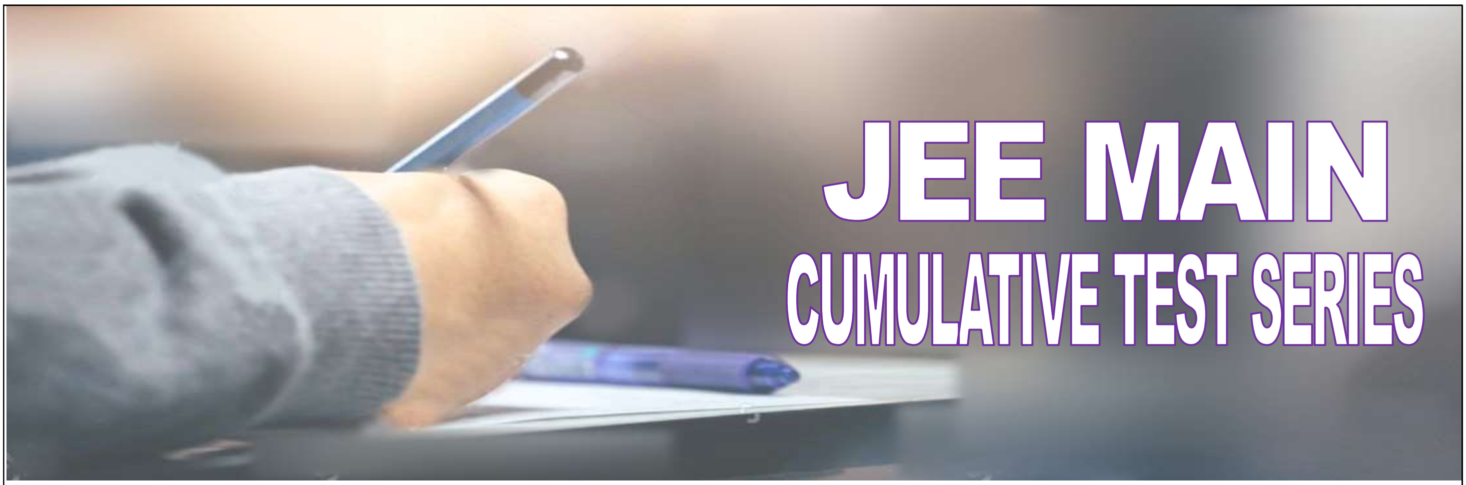



NARAYANA IIT ACADEMY INDIA

Sec: SR_*CO-SC(MODEL-B) Date:23-06-24
Time: 3 Hrs Max. Marks: 300

## 23-06-24_SR.IIT_STAR CO-SC(MODEL-B)_JEE MAIN_CTM-45_SYLLABUS

**PHYSICS:**

**PRESENT WEEK(60%): OPTICS-III :** Refraction at curved surface ,Lenses and combination of lenses
**MAINS EXPERIMENTS:** The focal length of Convex lens, using the parallax method.
**ADVANCED EXPERIMENTS:** focal length of a concave mirror and a convex lens using u-v method
**PREVIOUS WEEK(20%): OPTICS-II :** Refraction at plane surface, TIR and Prism, dispersion and deviation in thin prism
**MAINS EXPERIMENTS:** The plot of the angle of deviation vs angle of incidence for a triangular prism.
Refractive index of a glass slab using a travelling microscope.
**DESIGNATED TOPIC(20%):** Alternating Current

**CHEMISTRY:**

**PRESENT WEEK(60%):** Colour, Magnetic properties, Catalytic properties, alloy and interstitial compounds formation, oxides and oxoanions of metals, Preparation & Properties of Potassium dichromate, F-BLOCK
**PREVIOUS WEEK(20%): d-Block elements :** General Characteristics, electronic configuration, atomic & ionic size, density, ionisation enthalpy, melting points, enthalpy of atomization, Oxidation states, Chemical reactivity and E0 values, Nature of oxides
**DESIGNATED TOPIC(20%):** CHEMICAL & IONIC EQUILIBRIUM

**MATHEMATICS:**

**PRESENT WEEK(80%): Permutations & Combinations:** Fundamental Principles of counting, Linear permutations, Circular permutations
**PREVIOUS WEEK(20%):Binomial theorem :** Binomial theorem for a positive integral index, Constant term, Numerically greatest term, Properties of the Binomial coefficients, Multinomial Theorem
**DESIGNATED TOPIC(20%):** I) Triangular Inequalities(1qs) II)Rotation conics theorem(1qs) III) Geometrical interpretation(2qs) IV) Cube roots of unity(1qs) V) nth roots of unity(1qs)

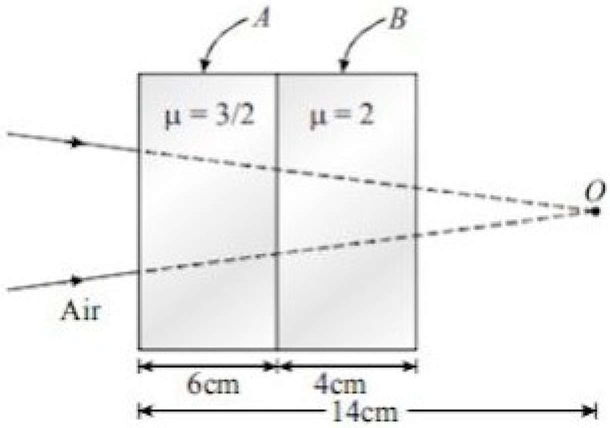
**PHYSICS** **MAX.MARKS: 100**

## SECTION – I
## (SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

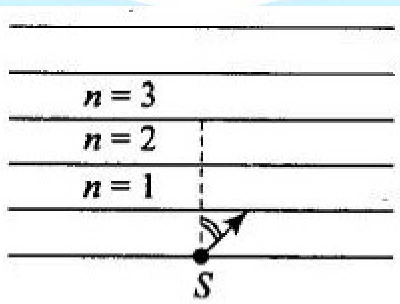
**Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.**

1. A convergent beam is incident on two slabs placed in contact as shown in Fig. Finally, the rays are converge at a distance (from left face of slab A is)




A)10 cm B) 18 cm C) 8 cm D) 6 cm

2. A point source S is placed at the bottom of different layers as shown in the figure. The refractive index of bottom most layer is $\mu_0$. The refractive index of any other upper layer is $\mu(n) = \mu_0 - \frac{\mu_0}{4n-18}$ *where* $n = 1, 2, .......$ A ray of light with angle I slightly more than $30^0$ starts from the source S. Total internal reflection takes place at the upper surface of a layer having n equal to




A) 3 B) 5 C) 4 D) 6

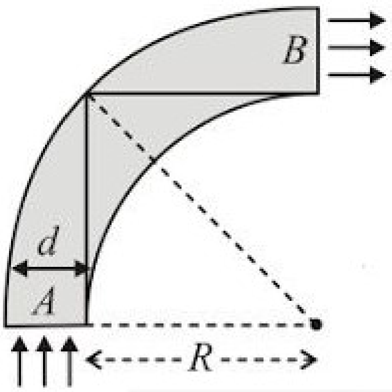
3. A prism of angle $80^0$ has refractive index $\sqrt{2}$. A light ray is incident at $45^0$ on one face. The total deviation of the ray is

A) $75^0$ B) $95^0$ C) $100^0$ D) $80^0$

4. A light ray is incident normally on one face of isosceles prism. It suffers two reflections inside prism and emerges normally out of the base. The base angles of prism are

A) $36^0, 36^0$ B) $72^0, 72^0$ C) $45^0, 45^0$ D) $60^0, 60^0$

5. A rod of glass $(\mu = 1.5)$ and of square cross section is bent into the shape shown I figure. A parallel beam of light falls on the plane flat surface A as shown in figure. If a is the width of a side and R is the radius of circular arc then for what maximum value of $\frac{d}{R}$ light entering the glass slab through surface A emerges from the glass through B




A) 1.5 B) 0.5 C) 1.3 D) None of these

6. A denser medium of refractive index 1.5 has a concave surface of radius of curvature $12cm$. A point object is situated in the denser medium at a distance of $9cm$ from the pole on the principal axis. Locate the image position due to refraction.

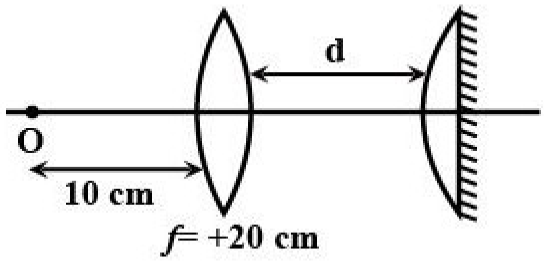
A) virtual image, at 4.8 cm from pole, inside the medium

B) virtual image, at 4.8cm from pole, inside the air

C) real image, at 9.6cm from pole inside the air

D) real image, at 9.6 cm from the inside the medium

7. A convex lens of focal length 20 cm and another plano-convex lens of focal length 40 cm are placed co-axially (see figure). The plano-convex lens is silvered on plane surface. What should be the distance d (in cm ) so that final image of the object 'O'is formed on O itself.




A) 10 B) 15 C) 20 D) 25

8. There is an equiconvex glass lens with radius of each face as R and $\left({}_a\mu_g = \frac{3}{2}\right)$. If there is water $\left({}_a\mu_w = \frac{4}{3}\right)$ in object space and air in image space, what is the focal length of the lens?

A) $\frac{5R}{2}$ B) $\frac{2R}{3}$ C) $\frac{3R}{2}$ D) $\frac{4R}{3}$

9. A plano convex lens of radius of curvature R and refractive index $\mu_1$ and a plano concave lens of same radius of curvature and refractive index $\mu_2$ are put in contact so that the combination appears as a parallel sided glass slab. What is the focal power of system?

A) $\frac{\mu_1 - \mu_2}{2R}$ B) $\frac{\mu_1 - \mu_2}{R}$ C) Infinity D) Zero

10. Two plano concave lenses of glass refractive index 1.5 have radii of curvature of 20cm and 30cm. if they are place in contact with curved surfaces towards each other and the space between them is filled with water $\left(\mu_w = \frac{4}{3}\right)$, find the focal length of the system.

A) $-50cm$ B) $95cm$ C) $-72cm$ D) $40cm$

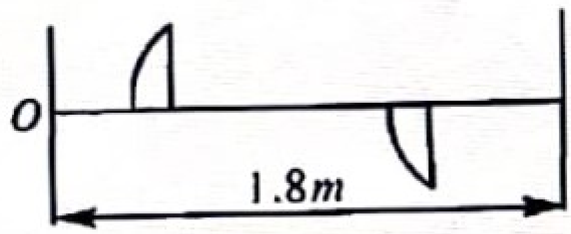
11. Two thin lenses when in contact produce a combination power $+10D$. When they are $0.25m$ apart. The power reduces to $+6D$. Find the focal lengths of the lenses.

A) $5cm, 50cm$ B) $10cm, 40cm$ C) $20cm, 30cm$ D) $12.5cm, 50cm$

12. A convex lens is placed somewhere in between an object and a screen. The distance between the object and the screen is 48cm. If the numerical value of the magnification produced by the lens is 3, focal length of the lens is

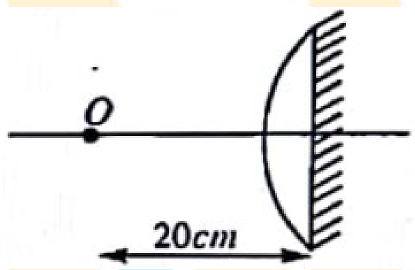
A) 6 cm B) 12 cm C) 24 cm D) 9 cm

13. A thin plano convex lens is split into two halves. One of the halves is shifted along the optical axis. The separation between object and image planes is $1.8m$. The magnification of the image formed by one half of lens is 2. Find the focal length of the lens and separation between the halves.




A) $0.3m, 0.2m$ B) $0.4m, 0.6m$ C) $0.1m, 0.6m$ D) $0.4m, 0.2m$

14. An object is placed at a distance of 20cm from a thin plano convex lens of focal length 15cm. The plane surface of lens is now silvered. What is the position of image?




A) 12cm to the left of lens system B) 20cm to the right of lens system
C) 12 cm to the right of lens system D) 20cm to the left of lens system

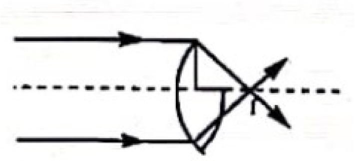
15. A light is incident on a transparent sphere of index $=\sqrt{2}$, at an angle of incidence $=45^0$. What is the deviation of a tiny fraction of the ray, which enters the sphere, undergoes two internal reflections, and then refracts out into air?

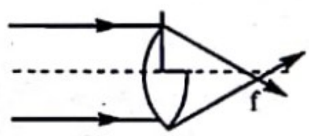
A) $270^0$ B) $240^0$ C) $120^0$ D) $180^0$

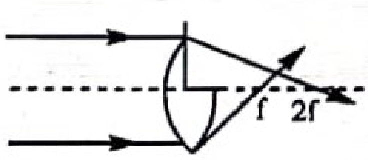
16. A plano convex lens has a thickness of 4 cm. When placed on a horizontal table with the curved surface in contact with it, the apparent depth of the bottom most point of the lens is found to be 3cm. If the lens is inverted such that the plane face is in contact with the table, the apparent depth of the centre of plane face is found to be 25/8 cm. Find the focal length of lens. Assume the thickness to be negligible while finding its focal length.

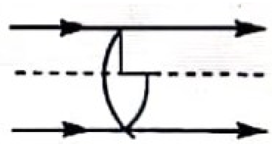
A) 75cm B) 60 cm C) 20 cm D) 100cm

17. Choose the correct ray diagram of a thin equi-convex lens which is cut from upper half as shown in the figure

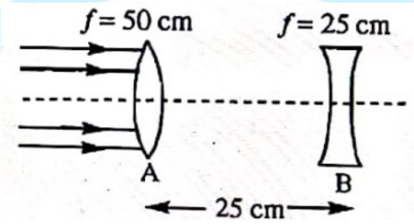
A) 


B) 


C) 


D) 

18. Two lenses shown are illuminated by a beam of parallel light from the left. Lens B is then moved slowly toward lens A. The beam emerging from lens B is



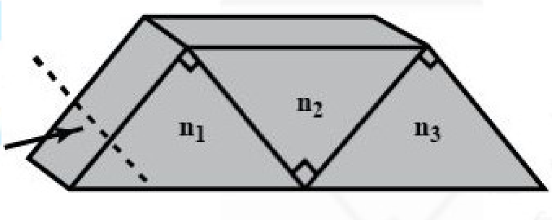

A) Always diverging

B) initially parallel and then diverging

C) Always parallel

D) initially converging and then parallel

19. Three right angled prisms of refractive indices $n_1$, $n_2$, $n_3$ are fixed together using an optical glue as shown in figure. If a ray passes through the prisms without suffering any deviation, then




A) $n_1=n_2=n_3$

B) $n_1=n_2\neq n_3$

C) $1+n_1=n_2+n_3$

D) $1+n_2^2=n_1^2+n_3^2$

20. A plane mirror is fixed at the bottom of a tank containing water. A small object is kept at height of 24 cm from the bottom and is viewed from a point vertically above it. The distance between the object and image in the mirror as 'appears' to the person is:

A) 48 cm

B) 64 cm

C) 36 cm

D) 54cm

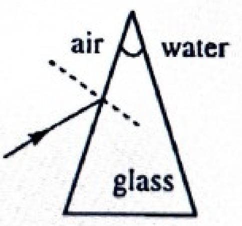
## SECTION-II
## (NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

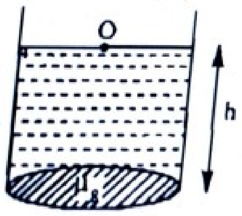
**Marking scheme: +4 for correct answer, -1 in all other cases.**

21. A circular disc of diameter d lies horizontally inside a metallic hemispherical bowl radius 8 cm. The disc is just visible to an eye looking over the edge. The bowl is now filled with a liquid of refractive index $\sqrt{3}$. Now, the whole of the disc just visible to the eye in the same position. The value of d (in cm)

22. Light travelling in air falls at an incident angle of $2^0$ on one refracting surface of a prism of refractive index 1.5 and angle of prism is $4^0$. The medium on the other side of second refracting surface is water $\left(\mu_w = \frac{4}{3}\right)$ as shown in diagram. The deviation produced by the prism in degree is




23. Bottom of a glass beaker is made up of a thin equi-convex lens having bottom side silvered as shown in the figure. Now water is filled in the beaker up to a height of h = 4m. The image of the point object floating at the middle of beaker on the surface of water coincides with itself. Find the radius of curvature of the lens $\left({}_a\mu_g = \frac{3}{2},\ {}_a\mu_w = \frac{4}{3}\right)$

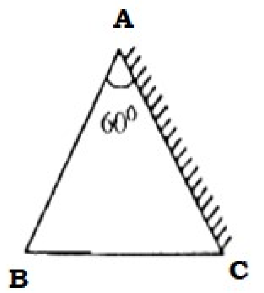



24. A beam of light falls on a glass plate $\left(\mu = \frac{3}{2}\right)$ of thickness 6.0 cm at an angle of $60^0$. Find the deflection of the beam on passing through the plate. (in cm).

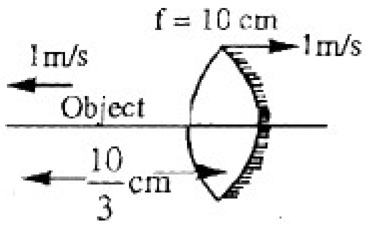
25. The flat bottom of cylinder tank is silvered and water $\left(\mu=\frac{4}{3}\right)$ is filled in the tank upto a height h. A small bird is hovering at a height 3h from the bottom of the tank. When a small hole is opened near the bottom of the tank, the water level falls at the rate of 1 cm/s. The bird will perceive that his velocity of image is 1/x cm/sec (in downward directions) where x is

26. A thin prism $P_1$ with angle $4^0$ and made for glass of refractive index 1.54 is combined with another thin prism $P_2$ made from glass refractive index 1.72 to produce dispersion without deviation. The angle of prism $P_2$ is

27. One of AC of an equilateral prism ABC is silvered as shown in fig. The angle of incidence of a light ray in order that it eventually leaves the prism in the opposite direction from base of prism is $n\times15^0$ $where\ 'n'\ is$ $\left(\mu=\sqrt{2}\right)$




28. One of the surfaces of a biconvex lens of focal length 10 cm silvered as shown in figure. Radis of curvature of silvered surface is 10 cm. At a given instant, if speed of object is 1 m/s, then the speed of image at that instant is $(10+\alpha)m/s.$ What will be the value of $\alpha$? $(\mu=1.5)$




29. The focal length of a thin biconvex lens is 20cm. When an object is moved from a distance 25cm in front of it 50cm, the magnitude of magnification of its image changes from $m_{25}$ to $m_{50}$. The ratio $\frac{m_{25}}{m_{50}}$ is

30. A thin biconvex lens of focal length 6.25cm is made of material of refractive index 1.5. It is into two identical pieces' perpendicular to its principal axis. One of the pieces is placed in water refractive index $\frac{4}{3}$. The focal power of the piece in water, in dioptre, is

**CHEMISTRY** **MAX.MARKS: 100**

## SECTION – I
## (SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

**Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.**

31. $I^-(aq) + MnO_4^-(aq) \xrightarrow{H^+} X + Mn^{++}(aq)$

$I^-(aq) + MnO_4^-(aq) \xrightarrow[\text{weaky OH}]{\text{natural or}} Y + MnO_2$

Products X, Y are respectively

A) $I_2, IO_3^-$ B) $IO_3^-, I_2$ C) $I_2, MnO_4^{2-}$ D) $IO_3^-, MnO_4^{2-}$

32. In general, actinoides exhibit more number of oxidation states than lanthanoids. This is because

A) There is a similarity between 4f and 5 f orbitals in their angular part of wave function

B) The actinoids are more reactive than lanthanoids

C) The 5f orbitals extended farther from the nucleus than the 4f-orbitals

D) The 5f orbitals are more buried than the 4f-orbitals

33. Fusion of $MnO_2$ with $KOH$ in presence of $O_2$ produces a salt W. alkaline solution of W upon a electrolytic oxidation yields another salt X . the manganese containing ions present in W and X, respectively are Y and Z. incorrect statement is

A) In both Y and Z, $\pi$-bonding occurs between p-orbitals of oxygen and d-orboitals of manganese

B) In aqueous acidic solution, Y undergoes dispro-portionation reaction to give Z and $MnO_2$

C) Both Y and Z are coloured and have tetrahedral shape

D) Y is diamagnetic in nature with Z is paramagnetic

34. Formation of interstitial compound makes the transition metal

A) More soft B) More ductile C) More metallic D) More hard

35. Arrange $(I)Ce^{3+}, (II)La^{3+}$ $(III)Pm^{3+}$ and $(IV)Yb^{3+}$ in increasing order of their ionic radii.

A) IV<III<I<II B) I<IV<III<IV C) IV<III<II<I D) III<II<I<IV

36. **Assertion :** The metals of 4d and 5 have greater enthalpies of atomization than the corresponding elements of the 3d series.

**Reason:** The metal –metal bond in 4d and 5d series are stronger than those in the 3d series

A) Both assertion and reason are true and the reason is the correct explanation of the assertion

B) Both assertion and reason are true but reason is not the correct explanation of the assertion

C) Assertion is true but reason is false.

D) Assertion is false but reason is true.

37. Match the property given in Column I with the element given in Column II

| | Column-I | | Column-II |
|---|---|---|---|
| i) | Lanthanoid wich shows +4 oxidation state | 1) | Pm |
| (ii) | Lanthanoid which can show +2 oxidation state | 2) | Ce |
| (iii) | Radioactive Lanthanoid | 3) | Eu |
| (iv) | Langhanoid which has $4f^7$ electronic configuration in +3 oxidation state | 4) | Gd |

| | i | ii | iii | iv | | i | ii | iii | iv |
|---|---|---|---|---|---|---|---|---|---|
| A) | 1 | 2 | 3 | 4 | B) | 2 | 3 | 1 | 4 |
| C) | 2 | 3 | 4 | 1 | D) | 2 | 4 | 3 | 1 |

38. Which one of the following alloys contains some of the lanthanoid metals?

A) Mischmetall B) Brass C) Bronze D) Ziggler-Natta

39. The ions from among the following which are colourless are:

(i) $Ti^{+4}$ (ii) $Cu^{+1}$ (iii) $Co^{+3}$ (iv) $Fe^{+2}$

A) (i) and (ii) only B) (i), (ii) and (iii)

C) (iii) and (iv) only D) (ii) and (iii) only

40. Which of the following ions has the maximum magnetic moment in aqueous solution?

A) $Mn^{+2}$ B) $Fe^{+2}$ C) $Co^{2+}$ D) $Cr^{+2}$

41. Among the following outermost configurations of transition metals, which shows the highest oxidation state.

A) $3d^3 4s^2$ B) $3d^5 4s^1$ C) $3d^5 4s^2$ D) $3d^6 4s^2$

42. Transition elements are used as catalyst because :

A) of high Ionic change B) of variable oxidation state

C) large surface area of reactants D) of their specific nature

43. Which of the following is arranged in order of increasing melting point?

A) $Zn < Cu < Ni < Fe$ B) $Fe < Ni < Cu < Zn$

C) $Ni < Fe < Zn < Cu$ D) $Cu < Zn < Fe < Ni$

44. Oxides of metal cation which is not amphoteric?

A) $Al^{3+}$ B) $Cr^{3+}$ C) $Fe^{3+}$ D) $Zn^{2+}$

45. $CrO_4^{2-}$ (yellow) changes to $Cr_2O_7^{2-}$ (orange) in Ph=x and vice-versa in pH=y, Hence, x and y are:

A) 6, 8 B) 6, 5 C) 8, 6 D) 7, 7

46. **Assertion :** Potassium dichromate gives deep red vapours with concentrated $H_2SO_4$ and sodium chloride.

**Reason:** The reaction of sodium chloride with solid $K_2Cr_2O_7$ and concentrated $H_2SO_4$ produces chromyl chloride.

A) Both assertion and reason are true and the reason is the correct explanation of the assertion

B) Both assertion and reason are true but reason is not the correct explanation of the assertion

C) Assertion is true but reason is false.

D) Assertion is false but reason is true.

47. The radius of $La^{3+}$ is 1.06 $\overset{0}{A}$, which of the following given values will be closed to the radius of $Lu^{3+}$ (At no. of $Lu = 71$, $La = 57$) –

A) 1.6 $\overset{0}{A}$ B) 1.4 $\overset{0}{A}$ C) 1.06 $\overset{0}{A}$ D) 0.85 $\overset{0}{A}$

48. $E^0$ value for couples $Cr^{+3}/Cr^{+2}$ and $Mn^{+3}/Mn^{+2}$ are -0.41 and +1.51 volts respectively. Considering these values select the correct option from the following statements:

A) $Cr^{+2}(aq)$ is more stable than $Cr^{+3}(aq)$

B) $Mn^{+3}(aq)$ is more stable than $Mn^{+2}(aq)$

C) $Cr^{+2}$ acts as a reducing agent and $Mn^{+3}$ acts as an oxidsing agent in their aqueous solutions.

D) $Mn^{+3}$ acts as a reducing agent and $Cr^{+2}$ acts as an oxidsing agent in their aqueous solutions.

49. The colour of $KMnO_4$ is due to:

A) $M \rightarrow L$ charge transfer transition B) d-d transition

C) $L \rightarrow M$ charge transfer transition D) $\sigma - \sigma^*$ transition

50. Reduction of the metal centre in aqueous permanganate ion involves

A) 1 electrons in neutral medium B) 5 electrons in neutral medium

C) 3 electrons in acidic medium D) 5 electrons in acidic medium

## SECTION-II
## (NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:
**Marking scheme: +4 for correct answer, -1 in all other cases.**

51. Identify the correct statements from following statements

(i) As the size decreases covalent character increases. Therefore $La_2O_3$ is more ionic and $Lu_2O_3$ is more covalent

(ii) The basic strength of hydroxides of lanthanoids decreases from $La(OH)_3$ to $Lu(OH)_3$

(iii) Stability of complexes increases as the size of lanthanoids decreases.

(iv) Radii of 4d and 5d block elements will be almost same.

(v) $Lu^{+3}, yb^{+2}, Ce^{+4}$ are diamagnetic

52. In $Cr_2O_7^{-2}$ ion the maximum number of bonds having the same $Cr-O$ bond length are____

53. Out of the following how many of them are coloured compounds:

$MnO_4^-, Cr_2O_7^{2-}, CrO_4^{2-}, Sc^{3+}, Ti^{4+}, Zn^{2+}, Mn^{3+}, Cu^{2+}, Fe^{2+}, Fe^{3+}$

54. Out of the following how many oxides are basic.

$TiO, Sc_2O_3, Ti_2O_3, VO, V_2O_5, CrO_3.Cr_2O_3, CuO.TiO_2$

55. What is the value of x in the Wilkinson's catalyst

$\left[RhCl(Ph_3P)_x\right]$ which is used as a homogenous catalyst in the hydrogenation of alkene

56. How many of the transition elements are called platinum metals?

57. Following is the process of preparing potassium dichromate.

(A) Chromite ore+potassium carbonate

Brown residue+Solution (B)

$\downarrow$ conc.$H_2SO_4$

Solution containing potassium dichromate (C)

What is the sum of oxidation state of Cr in A and B?

58. What is the sum $p\pi$-$d\pi$ bondings in manganate and permanganate ion

59. How many of the following pair of ions are either colourless or have almost same colour?

$(La^{3+}, Lu^{3+}); (Ce^{4+}, Yb^{2+}); (Sm^{3+}, Dy^{3+}); (Eu^{3+}, Tb^{3+})$

60. The general electronic configuration of actinoids are $5f^{1-14}6d^x7s^2$. How many values x can take.

## MATHEMATICS MAX.MARKS: 100

### SECTION – I
### (SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

**Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.**

61. The number of six letter words that can be formed using the letters of the word "ASSIST" in which S's alternate with other letters is

A)12 B)24 C)18 D)20

62. The number of 3 digit even numbers that can be formed by using the digits 0, 1, 2, 3, 4 and 5 is given by (repetition being not allowed)

A)50 B)52 C)48 D)60

63. Number of 3-digits numbers in which the digit at hundreds place is greater than the other two digits is

A)204 B)240 C)281 D)285

64. The number of ways in which 2 young men, 1 old and 1 lady can sit for a trip in a four-sitter car (two seats in the front and two in the back) so that the old man always sits in the back seat, it is being given that the two young men only can drive the car, is

A)8 B)12 C)16 D)14

65. The number of arrangements that can be formed with the letters of the word ORDINATE, so that the vowels occupy odd places is $k^2$ then the sum of the digits of $k=$

A)6 B)3 C)4 D)8

66. The number of arrangements which can be made out of the letters of the word ALGEBRA, so that no two vowels together is $360\lambda$ then $\lambda =$

A)2 B)4 C)18 (d)72

67. Two families with three members each and one family with four members are to be seated in a row. In how many ways can they be seated so that the same family members are not separated?

A)2! 3! 4! B)$(3!)^3\cdot(4!)$ C)$(3!)^2\cdot(4!)$ D)$3!(4!)^3$

68. The number of ways in which 5 boys and 3 girls can be seated on a round table if a particular boy $B_1$ and a particular girl $G_1$ never sit adjacent to each other, is

A)$5\times 6!$ B)$6\times 6!$ C)$7!$ D)$5\times 7!$

69. If seven women and seven men are to be seated around a circular table such that there is a man on either side of every woman, then the number of seating arrangements is

A) 6! 7! B) $(6!)^2$ C) $(7!)^2$ D) 7!

70. If $\alpha \in [-1,1]$ and $I(\alpha)$ denotes the term independent of $x$ in the expansion of $\left(x\sin^{-1}\alpha + \frac{1}{x}\cos^{-1}\alpha\right)^{10}$ then maximum value of $|I(\alpha)|$ is

A) ${}^{10}C_5 \frac{\pi^{10}}{2^{20}}$ B) ${}^{10}C_5 \frac{\pi^5}{2^{10}}$ C) ${}^{10}C_5 \frac{\pi^{10}}{2^5}$ D) ${}^{10}C_5 \left(\frac{1}{2}\right)^{10}$

71. If the $4^{th}$ term in the expansion of $\left(2+\frac{3x}{8}\right)^{10}$ has the maximum numerical value, then $x$ can lies in the interval(s)

A) $\left(-2, \frac{64}{21}\right)$ B) $\left(-\frac{60}{23}, -2\right)$ C) $\left(-\frac{64}{21}, -2\right) \cup \left(2, \frac{64}{21}\right)$ D) $\left(2, \frac{78}{23}\right)$

72. The coefficient of $x^4$ in the expansion of $\left(1+2x+\frac{3}{x^2}\right)^6$ is

A) 240 B) 250 C) 260 D) 230

73. The coefficient of $x^n$ in the polynomial $(x+{}^{2n+1}C_0)(x+{}^{2n+1}C_1)(x+{}^{2n+1}C_2)....(x+{}^{2n+1}C_n)$ is

A) $2^{n+1}$ B) $2^{n+1}-1$ C) $2^{2n}$ D) none of these

74. $\sum_{r=0}^{10} C_r \cdot \frac{2^{r+1}}{r+1} =$ (where $C_r = {}^{10}C_r$)

A) $\frac{3^{11}}{11}$ B) $\frac{2^{11}}{11}$ C) $\frac{3^{11}-1}{11}$ D) $\frac{2^{11}-1}{11}$

75. The value of $(2\cdot{}^1P_0 - 3\cdot{}^2P_1 + 2\cdot{}^3P_2 - ...$ up to 51th term) + (1!−2!+3!−.... upto 51th term) is equal to

A) 1 + (51)! B) 1 – 51(51)! C) 1 + (52)! D) 1

76. Let A = $\{x \mid x$ is a prime number and $x < 30\}$. The number of different rational numbers, whose numerator and denominator belong to A, is $13x$ then $x$ =

A) 7 B) 91 C) 72 D) 63

77. Let $n = 2005$. The least positive integer $k$ for which $k(n^2)(n^2-1^2)(n^2-2^2)(n^2-3^2)........(n^2(n-1)^2) = r!$ some positive integer $r$ is

A) 4 B) 8 C) 2 D) 6

78. If α is non-real and $\alpha = \sqrt[5]{1}$,, then the value of $2^{|1+\alpha+\alpha^2+\alpha^{-2}-\alpha^{-1}|}$ is equal to

A) 4 B) 2 C) 1 D) none of these

79. If $z^2 - z + 1 = 0$, then $z^n - z^{-n}$, where n is a multiple of 3, is

A) $2(-1)^n$ B) 0 C) $(-1)^{n+1}$ D) None of these

80. The complex numbers $z_1, z_2$ and $z_3$ satisfying $\frac{z_1 - z_3}{z_2 - z_3} = \frac{1 - i\sqrt{3}}{2}$ are the vertices of a triangle which is

A) of area zero B) right angled isosceles

C) equilateral D) obtuse angled isosceles

## SECTION-II
## (NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

**Marking scheme: +4 for correct answer, -1 in all other cases.**

81. An n-digit number is a positive number with exactly $n$ digits. Nine hundred distinct $n$-digit numbers are to be formed using only the three digits 2, 5 and 7. The smallest value of $n$ for which this is possible is ________

82. The number of three digit numbers having only two consecutive digits identical is $abc$ then $a+b+c$ is ________

83. The number of ways of arranging the letters of the word NALGONDA, such that the letters of the word GOD occur in that order (G before and O and O before D), is P then P/420

84. If the letters of the word 'MOTHER' be permuted and all the words so formed (with or without meaning) be listed as in a dictionary, then the position of the word 'MOTHER' is ________

85. The number of rational terms in the expansion of $(\sqrt{2} + \sqrt[3]{3} + \sqrt[6]{5})^{10}$ is ________

86. The tens digit of 1! + 2! + 3!+.....+ 49! is ________

87. The number of positive terms in the sequence $x_n = \frac{195}{4 \cdot {}^nP_n} - \frac{{}^{n+3}P_3}{{}^{n+1}P_{n+1}}, n \in N$ is ________

88. The number of points of intersection of $|z-(4+3i)| = 2$ and $|Z| + |z-4| = 6, z \in C$, is

89. Let $S = \{z \in C : z^2 + \bar{z} = 0\}$. Then $\sum_{z \in S} (\mathrm{Re}(z) + \mathrm{Im}(z))$ is equal to

90. If for the complex number z satisfying $|z - 2 - 2i| \le 1$, the maximum value of $|3iz + 6|$ is attained at a + ib, then a + b is equal to